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A Report on
Brain Tumor Detection and MRI Analysis using
Traditional Computer Vision Techniques

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INTRODUCTION

Brain tumors are abnormal growths of cells inside the brain that can affect normal brain functioning and may lead to severe neurological complications if not detected at an early stage. These tumors can be either benign (non-cancerous) or malignant (cancerous), and their accurate detection is extremely important for proper medical diagnosis and treatment planning. Medical imaging techniques such as Magnetic Resonance Imaging (MRI) are widely used by doctors and radiologists because MRI provides highly detailed images of brain tissues without exposing patients to harmful radiation.

In recent years, Artificial Intelligence and Deep Learning techniques such as Convolutional Neural Networks (CNNs) have become popular for medical image analysis. However, traditional computer vision techniques still remain highly relevant because they are computationally efficient, easier to interpret, and suitable for systems where large computational resources are unavailable. Traditional computer vision approaches also provide better explainability because each processing stage can be visualized and understood individually.

This project focuses on the development of a Brain Tumor Detection and MRI Analysis System using traditional computer vision techniques instead of deep learning models. The project utilizes image preprocessing, frequency-domain image enhancement, image segmentation, morphological operations, feature extraction, and machine learning-based classification to detect whether a tumor is present in MRI brain scans.

The project also includes advanced functionalities such as:

- Tumor severity estimation
- Tumor area percentage analysis
- Heatmap visualization of suspicious regions
- Interactive Streamlit-based graphical user interface
- Automated PDF report generation

The entire system is implemented using Python, OpenCV, Scikit-image, Scikit-learn, and Streamlit. The project demonstrates how classical computer vision techniques can still be effectively applied in medical imaging applications while also covering important syllabus topics such as image enhancement, segmentation, and morphological operations.



OBJECTIVE

The main objective of this project is to design and implement a Brain Tumor Detection System using traditional computer vision techniques for MRI image analysis. The project aims to classify MRI brain scans into two categories:

- Tumor
- No Tumor

The specific objectives of the project are as follows:

1. To preprocess MRI brain images for improving image quality and reducing computational complexity.
2. To implement image enhancement using frequency-domain processing techniques such as Fourier Transform.
3. To perform image segmentation using thresholding methods in order to isolate suspicious tumor regions.
4. To apply morphological operations for noise removal and improvement of segmented tumor regions.
5. To extract important image features such as:
 - Area
 - Perimeter
 - Circularity
 - Mean Intensity
 - Extent
6. To classify MRI images using the K-Nearest Neighbors (KNN) machine learning algorithm.
7. To estimate tumor severity levels based on segmented tumor size.
8. To calculate the percentage of affected tumor area in the MRI image.
9. To generate heatmap overlays for improved visualization and interpretability.
10. To build an interactive Streamlit-based web application for real-time MRI analysis.
11. To generate downloadable PDF reports containing analysis results and extracted information.

The project is developed with the intention of demonstrating the practical application of traditional computer vision techniques in medical image analysis while maintaining simplicity, interpretability, and educational value.



SYSTEM ARCHITECTURE

3.1 Dataset Used

The project uses the Brain Tumor MRI Dataset obtained from Kaggle. The dataset contains MRI brain scans categorized into the following classes:

- Glioma
- Meningioma
- Pituitary Tumor
- No Tumor

For simplification and implementation using traditional computer vision methods, the dataset was converted into binary classification:

- Tumor
- No Tumor

All tumor classes (Glioma, Meningioma, and Pituitary) were merged into a single “Tumor” category. The dataset contains both training and testing images and includes MRI scans with different tumor sizes, shapes, and intensity patterns, making it suitable for evaluating segmentation and feature extraction techniques.

3.2 System Architecture

The complete system architecture consists of the following stages:

1. MRI Image Upload
2. Image Preprocessing
3. Frequency-Domain Enhancement
4. Image Segmentation
5. Morphological Processing
6. Feature Extraction
7. KNN Classification
8. Tumor Severity Estimation
9. Tumor Area Percentage Calculation
10. Heatmap Visualization



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11. Result Display in Streamlit UI
12. PDF Report Generation

The architecture follows a sequential image-processing pipeline in which each stage improves the output generated by the previous stage.

3.3 Image Preprocessing

Image preprocessing is the first stage of the system. MRI images obtained from the dataset may have different dimensions and intensity distributions. Therefore, preprocessing is required to standardize the images before further analysis.

The preprocessing steps include:

- Converting the image to grayscale
- Resizing the image to 256×256 pixels
- Noise reduction using Gaussian Blur

Grayscale conversion reduces computational complexity and simplifies image analysis because only intensity information is required for segmentation and feature extraction.

Image resizing ensures that all MRI images have a uniform size, making feature extraction consistent across the dataset.

Gaussian Blur is applied to reduce noise and smooth unwanted intensity variations before segmentation.

Libraries used:

- OpenCV
- NumPy



3.4 Frequency Domain Image Enhancement

The project implements image enhancement in the frequency domain using Fourier Transform techniques. This module directly covers the syllabus topic:

“Image Enhancement in Frequency Domain.”

The enhancement process is performed using:

- Discrete Fourier Transform (DFT)
- Frequency shifting
- High-pass filtering

The purpose of frequency-domain enhancement is to improve high-frequency structures such as edges and tumor boundaries. A high-pass filter suppresses low-frequency components and highlights detailed structures present in MRI images.

The following steps are performed:

1. Apply DFT to the MRI image.
2. Shift low-frequency components to the center.
3. Create a high-pass mask.
4. Filter frequency components.
5. Reconstruct the enhanced image using inverse DFT.

This enhancement improves the visibility of suspicious tumor regions and helps achieve better segmentation results.

3.5 Image Segmentation

Segmentation is one of the most important stages of the project. The purpose of segmentation is to separate suspicious tumor regions from the background brain tissues.

The project uses Otsu Thresholding for segmentation.

Otsu Thresholding automatically determines an optimal threshold value based on image intensity distribution. The method converts the enhanced MRI image into a binary image where:

- White regions represent possible tumor areas
- Black regions represent background regions



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The segmentation process includes:

1. Applying Gaussian smoothing
2. Computing threshold value using Otsu's algorithm
3. Generating binary segmented image

The segmented output acts as the base for further morphological processing and feature extraction.

3.6 Morphological Operations

Morphological operations are applied to improve the segmented image by removing noise and refining the tumor region boundaries.

This section directly covers the syllabus topic:
“Morphological Operations on Image.”

The following operations are used:

OPENING

Opening removes small noise particles from the segmented image.

CLOSING

Closing fills small holes inside segmented tumor regions.

DILATION

Dilation enlarges the segmented tumor region slightly and improves contour visibility.

A 3×3 structuring kernel is used for performing these operations.

Morphological processing significantly improves the quality of the segmented mask and helps produce better feature extraction results.



3.7 Feature Extraction

After segmentation and morphological processing, important image features are extracted from the segmented tumor region.

The extracted features include:

Feature	Description
Area	Total number of tumor pixels
Perimeter	Length of tumor boundary
Circularity	Shape compactness of tumor
Mean Intensity	Average brightness value
Extent	Ratio of region area to bounding box area

These features help represent the structural characteristics of tumor regions numerically.

Feature extraction plays an important role because the KNN classifier uses these extracted values as input for prediction.

3.8 KNN Classification

The project uses the K-Nearest Neighbors (KNN) algorithm for classification.

KNN is a supervised machine learning algorithm that classifies images based on similarity between extracted feature vectors.

The algorithm works as follows:

1. Extract features from training images.
2. Store feature vectors and labels.
3. Compute distances between test image features and training features.
4. Identify nearest neighbors.
5. Predict class based on majority voting.



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The classifier predicts:

- Tumor
- No Tumor

KNN was selected because:

- Simple implementation
- Suitable for low-dimensional handcrafted features
- Easy to interpret
- Computationally efficient for educational projects

The model achieved an approximate accuracy of 80–82% on the testing dataset.

3.9 Tumor Severity Estimation

An additional feature implemented in the project is tumor severity estimation.

The segmented tumor area is analyzed and categorized into:

- Small Tumor
- Medium Tumor
- Large Tumor

Severity estimation is based on the extracted tumor area feature.

This functionality improves the usefulness of the system by providing additional quantitative analysis instead of only binary prediction.

3.10 Tumor Area Percentage Estimation

The system estimates the approximate percentage of affected MRI area occupied by the tumor.

The percentage is calculated using:

$$\text{Tumor Area Percentage} = (\text{Tumor Pixels} / \text{Total Pixels}) \times 100$$



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This feature provides additional information regarding the spread of the tumor region within the MRI image.

The calculated percentage is displayed in the user interface along with prediction results.

3.11 Heatmap Overlay Visualization

To improve visualization and interpretability, the project generates a heatmap overlay highlighting suspicious tumor regions.

The segmented mask is converted into a colored heatmap using OpenCV color-mapping techniques. The heatmap is then superimposed over the original MRI image.

The heatmap overlay:

- Improves visual understanding
- Highlights suspicious regions
- Makes the system appear similar to modern medical imaging tools

This feature enhances the presentation and usability of the system.

3.12 Streamlit-Based User Interface

A Streamlit-based web application is developed to provide an interactive graphical user interface.

The interface allows users to:

- Upload MRI images
- View processing stages
- Observe segmentation results
- Analyze tumor severity
- View tumor area percentage
- Visualize heatmaps
- Download PDF reports

The Streamlit UI improves user interaction and makes the project suitable for real-time demonstrations and presentations.



3.13 PDF Report Generation

The project also includes automatic PDF report generation functionality.

The generated report contains:

- MRI prediction result
- Confidence score
- Tumor severity
- Tumor area percentage
- Extracted features
- Visualization results

This feature makes the project more practical and professional.

RESULTS

Experimental Results

The developed system successfully performs:

- MRI enhancement
- Tumor segmentation
- Morphological refinement
- Feature extraction
- KNN classification
- Tumor severity estimation
- Heatmap generation
- PDF report generation

The Streamlit interface provides real-time MRI analysis and visualization.

The system achieved approximately:

- Accuracy: 81–82%



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The confusion matrix obtained from the testing dataset demonstrates that the classifier performs well on structured MRI images.

The project successfully demonstrates that traditional computer vision techniques can effectively perform brain tumor detection without using deep learning models.

(Insert screenshots here)

1. Original MRI
2. Enhanced MRI
3. Segmented Mask
4. Boundary Detection
5. Heatmap Overlay
6. Streamlit UI
7. PDF Report Output

CONCLUSION

The project successfully implements a Brain Tumor Detection and MRI Analysis System using traditional computer vision techniques.

The system integrates multiple image-processing stages including:

- Frequency-domain image enhancement
- Segmentation
- Morphological operations
- Feature extraction
- KNN classification

The project demonstrates that traditional computer vision methods can still provide effective solutions for medical image analysis while remaining computationally efficient and interpretable.

Additional functionalities such as:

- Tumor severity estimation
- Tumor area percentage analysis
- Heatmap visualization



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- Interactive Streamlit interface
- PDF report generation

make the project more advanced, user-friendly, and visually impressive.

The project also aligns strongly with important computer vision syllabus concepts such as image enhancement, segmentation, and morphology.

Overall, the system provides a practical demonstration of how classical computer vision techniques can be applied to real-world medical imaging problems.

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